BSCE Esoft Quiz-Part2

Question #: 1

A 24-year-old man comes to the physician for a routine examination. During auscultation of the chest for breath sounds, the physician starts by identifying the individual intercostal spaces by finding the sternal angle, which is located at the level of the 2nd costal cartilages, and counting downwards. This anatomical landmark most likely lies at the level of which of the following intervertebral discs?

A. T1-2
B. T2-3
C. T3-4
✓D. T4-5
E. T5-6

Rationale: The sternal angle of Louis is the anterior boundary of the plane of Ludwig which spans from the sternal angle anteriorly to the T4-T5 intervertebral disc posteriorly.

Question #: 2

A 1-week-old girl is brought to the physician by her mother because of a 2-day history of a liquid discharge from a small opening in her umbilicus. Contrast material is introduced into the opening, and a radiograph is taken which shows that the opening is continuous with the bladder. Which of the following is the most likely diagnosis?

A. Urachal sinus
✓B. Urachal fistula
C. Urachal cyst
D. Ectopic ureter
E. Exstrophy of the bladder

Rationale: The urachus is the fibrous remnant of the allantois, a canal that drains the urinary bladder of the fetus. It extends from the apex of the bladder to the umbilicus. If the urachal lumen fails to close, you may have several remnants: urachal cyst - the proximal and distal ends of the lumen close, leaving a cyst in the middle, urachal sinus - the proximal or distal end of the lumen fail to close leaving a sinus that opens into the umbilicus or into the bladder, urachal fistula - the entire urachus remains patent leaving a connection between the bladder and the umbilicus.

Question #: 3

A 59-year-old woman comes to the physician for a routine examination. Auscultation shows a systolic murmur which is loudest at the second intercostal space on the right of the sternum. Echocardiography shows valvular stenosis. Which of the following structures most likely receives the blood that is being ejected through the narrowed opening?

A. Left ventricle
B. Right ventricle
✓C. Aorta
D. Pulmonary trunk
E. Left atrium

Rationale: The valve that can be auscultated at the right 2nd intercostal space is the aortic valve. The aortic valve communicates between the left ventricle and the aorta

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Question #: 4

A 32-year-old woman comes to the emergency department because of a 2-hour history of heavy vaginal bleeding. She is 18 weeks pregnant. Despite efforts to save the fetus she aborts spontaneously. Necropsy shows severe defects of the indicated structure including stenosis and malrotation. Abnormal development of which of the following would most likely cause these defects?

A. Dorsal aorta
B. Left 4th arch artery
C. Left 6th arch artery
D. Right 4th arch artery
E. Right 6th arch artery
✓F. Truncus arteriosus
G. Bulbus cordis

Rationale: Correct answer F: Truncus arteriosus is the correct answer. The marked structure is ascending aorta. During development, the truncus arteriosus is divided by a spiral shaped aorto-pulmonary septum into two vessels, the ascending aorta, and the pulmonary trunk.

Incorrect answer: Dorsal aorta is incorrect; The paired dorsal aortae arise from aortic arches that in turn arise from the aortic sac. The primary dorsal aorta is located deep to the lateral plate of mesoderm and move from lateral to medial position with development and eventually will fuse with the other dorsal aorta to form the descending aorta.

Left 4th aortic arch is incorrect. The left arch of the fourth pair forms the segment of normal left aortic arch between the left common carotid and subclavian arteries.

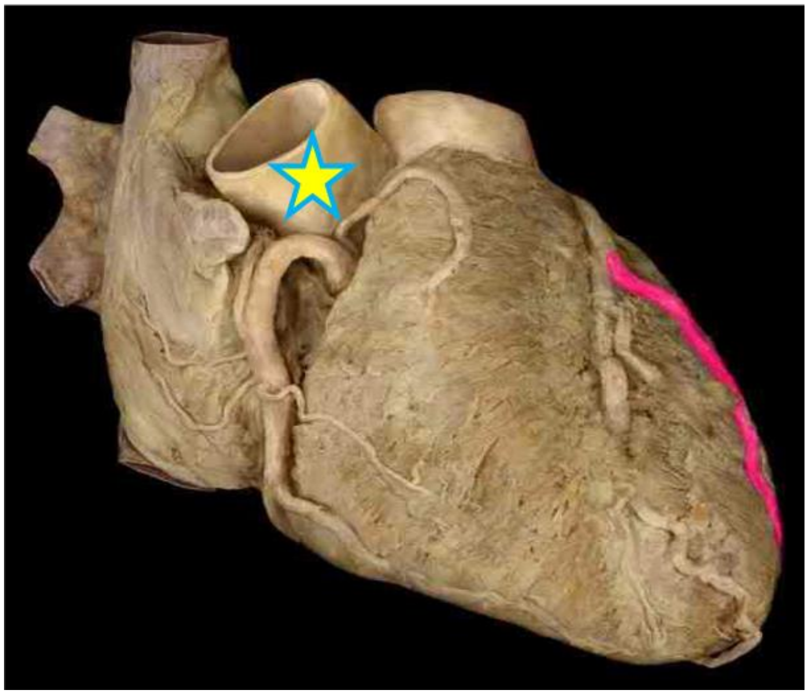
Left 6th arch artery is incorrect. The left arch of the sixth pair contributes to the formation of the main and left pulmonary arteries and ductus arteriosus; this duct obliterates a few days after birth.

Right 4th arch artery is incorrect. The right fourth arch forms the proximal right subclavian artery. The distal right subclavian artery is derived from a portion of the right dorsal aorta and the right seventh intersegmental artery.

Right 6th arch artery is incorrect. The proximal part of the sixth right arch persists as the proximal part of the right pulmonary artery while the distal section degenerates.

Bulbus cordis is incorrect, bulbus cordis gives rise to the outflow tract (smooth part) of the ventricles of the heart. The superior end of the bulbus cordis is also called the conotruncus.

Attachment:



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Question #: 5

A 45-year-old man is brought to the emergency department with tightening pain in his chest. He has a past history of two myocardial infarcts with stent placement. Physical examination show profuse sweating and a pulse rate of 120 bpm. Cardiac enzymes are elevated. ECG results indicate ischemia of the diaphragmatic surface of the ventricles and posterior part of the interventricular septum. During coronary artery bypass, the surgeon is careful to avoid damaging the vein that accompanies the blocked vessel. Which of the following veins accompany the affected artery?

✓A. Middle cardiac
B. Small cardiac
C. Coronary sinus
D. Anterior cardiac
E. Great cardiac

Rationale: Correct answer. Middle cardiac is the correct answer. The damaged artery during the surgery in this patient is posterior interventricular artery. The vein that accompanies this artery is the middle cardiac vein.
Incorrect answer. Small cardiac vein is incorrect. This artery is located in the grove between the right atrium and right ventricle.
Coronary sinus is incorrect. It is located in the posterior part of the atrio-ventricular groove and empties into the right atrium.
Anterior cardiac vein is incorrect. These are 3 or 4 in number and drain from infundibulum of the right ventricle.
Great cardiac vein is incorrect. It is located in the anterior interventricular grove along with anterior interventricular artery.

Question #: 6

During a heart transplant procedure, the surgeon inserted his left index finger through the transverse pericardial sinus and then pulled forward on the two large vessels lying ventral to his finger. Which vessels were these?

A. Pulmonary trunk and brachiocephalic trunk
✓B. Pulmonary trunk and aorta
C. Pulmonary trunk and superior vena cava
D. Superior vena cava and aorta
E. Superior vena cava and right pulmonary artery

Rationale: Correct answer. Pulmonary trunk and aorta is the correct answer. Transverse pericardial sinus is clinically important because passing one end of clamp through the sinus, and the other end anterior to the aorta/pulmonary trunk will allow complete blockage of blood output. This is done during some heart surgeries. This sinus is present anterior to the superior vena cava and posterior to the aorta and pulmonary trunk.
Incorrect answer. Pulmonary trunk and brachiocephalic trunk is incorrect. Transverse sinus is present anterior to the superior vena cava and posterior to the aorta and pulmonary trunk.
Pulmonary trunk and superior vena cava, Superior vena cava and aorta, Superior vena cava and right pulmonary artery are incorrect. Transverse sinus is present anterior to the superior vena cava and posterior to the aorta and pulmonary trunk.

Question #: 7

A 57-year-old woman is admitted to the hospital with severe pain radiating from the right lumbar region of the abdomen to the pubic symphysis on the left. Ureteric obstruction is suspected and confirmed by an intravenous pyeloureterogram which shows hydronephrosis of the left kidney and part of the left ureter as seen in the attached image. Which of the following is the most likely location of the urinary stone?

A. In the minor calyx
B. In the urethra
✓C. Pelvic brim/where crosses the common iliac artery
D. At the ureterovesical junction
E. At the ureteropelvic junction

Rationale: Correct C: In the pyeloureterogram, contrast can be seen on the right side, showing the right kidney, the entire extent of the right ureter including its entrance into the bladder. On the left, there is hydronephrosis of the left kidney and part of the left ureter down to the pelvic brim/where it crosses the common iliac artery. There is no contrast beyond this point. This means that the stone is lodged at this point.
Incorrect: If the stone were lodged in the minor calyx, this will be the only location where contrast would be absent, but the rest of the kidney and ureters would still have contrast. This imaging technique does not show the urethra. If the blockage were at the ureteropelvic junction, there would be hydronephrosis of the left kidney and renal pelvis, but none in the left ureter. If the blockage were at the ureterovesical junction, there would be hydronephrosis in the left kidney and the entire left ureter from its entrance into the bladder.

Attachment:

Ureteric obstruction.jpg

Question #: 8

A 40-year-old man is brought to the emergency department because of a 2-hour history of progressive shortness of breath, chest pain, and palpitations. Imaging confirms a pulmonary embolism, and he is started on intravenous unfractionated heparin, which prolongs the intrinsic coagulation cascade. Which of the following laboratory tests is most appropriate to monitor the activity of this pathway in this patient?

A. Bleeding time
B. Platelet count
C. Prothrombin time
✓D. Activated partial thromboplastin time
E. Ristocetin cofactor assay

Rationale: The test for the intrinsic clotting pathway is APTT. Prothrombin time (PT or INR) assesses the extrinsic coagulation pathway. Bleeding time is an indicator of platelet aggregation and platelet function.
Ristocetin cofactor assay is specific for von Willebrand factor deficiency.

Question #: 9

A 25-year-old woman comes to the physician because of a 2-day history of shortness of breath. She has a 5-year history of asthma, with symptoms triggered by allergens and exercise. She is prescribed an inhaled corticosteroid to decrease eicosanoid synthesis. Which of the following enzymes is most likely inhibited by this drug?

A. Thromboxane synthase
B. Phospholipase C
C. Lipoxygenase
✓D. Phospholipase A2
E. Cyclooxygenase 1
F. NADPH oxidase
G. Myeloperoxidase

Rationale: Cortisol in pharmacologic doses inhibits phospholipase A2, which releases arachidonic acid from membrane phospholipid. Arachidonic acid is the precursor of all the eicosanoids (prostaglandins, thromboxane, and leukotrienes). Cortisol also inhibits COX-2 (induced by inflammation).
NSAIDs and aspirin are inhibitors of COX-1 and COX-2.
Lipoxygenase inhibitors specifically inhibit the formation of leukotrienes.

Question #: 10

A 3-year-old girl is brought to the physician by her mother because of a 3-month history of progressive weakness and fatigue. She has no history of recent infections or medication use. Physical examination shows mild pallor and slight jaundice. Laboratory studies show:

- Hemolytic anemia
- Elevated 2,3-bisphosphoglycerate (2,3-BPG) levels in red blood cells
- Peripheral smear shows echinocytes (burr cells)

Which of the following disorders is the most likely diagnosis?

A. Glucose 6-phosphate dehydrogenase deficiency
B. Alpha thalassemia
✓C. Pyruvate kinase deficiency
D. Sickle cell anemia
E. HbC disease
F. Beta thalassemia

Rationale: The vignette describes hemolytic anemia due to the inability to maintain ionic balance in RBCs. Deficiency of glycolytic enzymes leads to reduced ATP formation. As a result, the membrane Na-K pumps are not as efficient. The intracellular Na accumulation leads to osmotic lysis. In pyruvate kinase deficiency, 2,3-BPG levels are elevated, whereas in hexokinase deficiency, 2,3-BPG levels are low.
In patients with G6PD deficiency (X-linked), males are typically affected. Hemolysis is due to the accumulation of reactive oxygen species due to a deficiency of NADPH.
Hemolysis in beta thalassemia is due to the accumulation of excess alpha-globin chains, which damage RBC membranes.
In HbS, polymerization of HbS results in crescent-shaped RBCs removed by the spleen.
In HbC disease, HbC is less soluble than HbA and reduces the lifespan of RBC.
In alpha-thalassemia, RBC lifespan is reduced due to the accumulation of abnormal hemoglobins like HbH (four beta chains).

Question #: 11

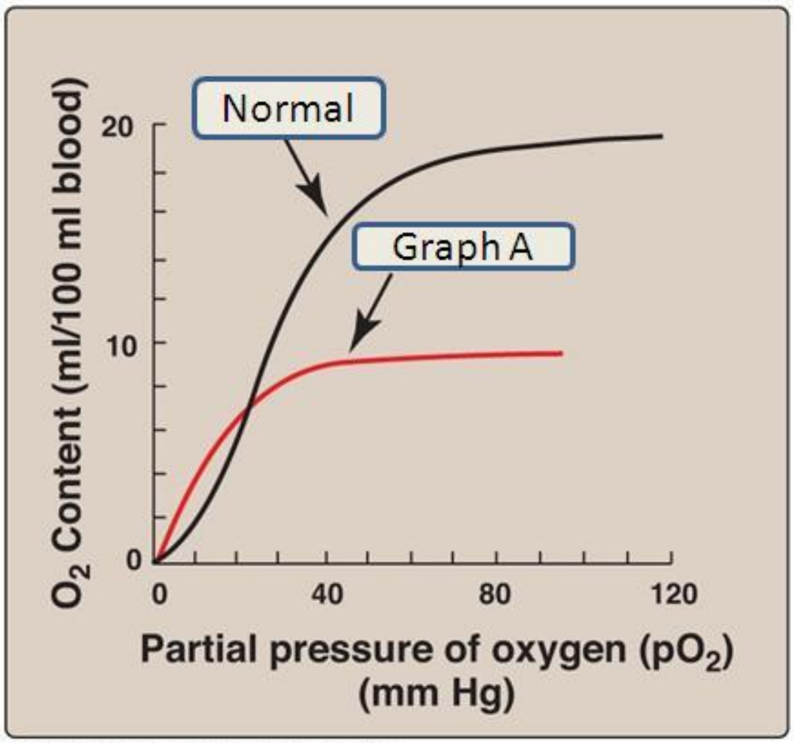
An investigator is studying the oxygen-hemoglobin dissociation curve in a patient. The curve from a normal volunteer is represented by the black line in the graph and labeled “Normal.” The patient’s oxygen dissociation curve is labeled Graph A. Which of the following conditions is most likely to produce the oxygen dissociation curve represented by Graph A?

A. Presence of fetal hemoglobin
B. Pyruvate kinase deficiency
C. Adaptation to high altitude
✓D. Presence of carbon monoxide
E. Hypothermia
F. Beta-thalassemia

G. Hexokinase deficiency

Rationale: Carbon monoxide binds to hemoglobin at the oxygen-binding site and effectively reduces the number of available oxygen-binding sites on hemoglobin. Also, the binding of carbon monoxide to one of the subunits of hemoglobin prevents the unloading of oxygen from the other binding sites. CO stabilizes the R state of hemoglobin. Fetal hemoglobin (HbF) has a higher affinity to oxygen and shifts the oxygen dissociation curve to the left. Pyruvate kinase deficiency shifts the ODC to the right due to higher levels of 2,3-BPG. Hexokinase deficiency shifts the ODC to the left due to lower levels of 2,3-BPG. In a person adapted to high altitude, 2,3-BPG levels increase and cause a right shift of the ODC.

Attachment:



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Question #: 12

A 20-year-old woman comes to the physician for advice regarding weight loss. She reports difficulty controlling her appetite despite dietary changes. Her waist circumference is 95 cm (37.5 in); BMI is 29 kg/m². Laboratory studies show elevated serum leptin concentration. Which of the following best describes the physiological function of this adipokine?

A. Increases appetite
✓B. Suppresses appetite
C. Decreases energy expenditure
D. Decreases satiety
E. Causes negative nitrogen balance

Rationale: Leptin is an adipokine (hormone released from the adipose tissues). There is increased leptin secretion when a person gains weight. High leptin levels suppress appetite and signal satiety. Leptin also increases energy expenditure.

Ghrelin, released from the stomach, signals hunger.

Question #: 13

An investigator is studying intracellular signal transduction pathways in cultured cells. Administration of drug X causes activation of protein kinase C (PKC) in the target cells. Which of the following molecules most likely mediates this effect on the protein?

✓A. Diacylglycerol
B. Inositol 1, 4, 5-tris-phosphate
C. Phosphatidylinositol 4,5-bisphosphate
D. Cyclic-AMP
E. Cyclic-GMP
F. Nitric oxide
G. Calmodulin

Rationale: Protein kinase C is activated by increased intracellular levels of DAG and Calcium ions.
IP3 increases the efflux of Calcium from the endoplasmic reticulum. IP3 indirectly activates protein kinase C by increasing intracellular calcium concentrations. Calcium activates protein kinase C.
cAMP increases the activity of protein kinase A.
NO increases cGMP levels, which in turn activate Protein kinase G.

Question #: 14

A 29-year-old man comes to the emergency department because of a 10-hour history of severe diarrhea and vomiting. He reports that his symptoms began after a weekend hiking trip during which he consumed wild mushrooms. A provisional diagnosis of “death cap” mushroom poisoning is made. Which of the following proteins is most likely inhibited by the toxin in this mushroom?

✓A. RNA polymerase-II
B. Peptidyl transferase
C. DNA gyrase
D. Eukaryotic elongation factor-2 (eEF-2)
E. Adenylate cyclase

Rationale: Amanitin (Amanita phalloides) inhibits eukaryotic transcription by inhibiting RNA polymerase II (required for mRNA synthesis).

Cycloheximide inhibits eukaryotic peptidyl transferase activity in translation.
The eEF-2 is inhibited by diphtheria toxin. The diphtheria toxin causes ADP-ribosylation of eEF-2 and inhibits translation (elongation of translation).
DNA gyrase inhibitors are quinolones, eg, Ciprofloxacin.

Question #: 15

A 42-year-old woman comes to the physician because of a 6-day history of worsening cough and fever. She reports shortness of breath and general malaise. Her respirations are increased. Auscultation of the lungs shows diffuse crackles. Chest imaging is consistent with a lower respiratory tract infection. Sputum culture identifies erythromycin-sensitive bacteria, and she is started on intravenous erythromycin. Which of the following bacterial processes is most likely inhibited by this drug?

A. Binding of aminoacyl tRNA to the ribosome
B. Unwinding of supercoiled DNA
C. Initiation of translation and the ribosome assembly
✓D. Translocation of peptidyl-tRNA from the A-site to the P-site
E. Transcription and RNA synthesis
F. Peptidyl transferase activity of ribosome

Rationale: Erythromycin inhibits protein synthesis (elongation phase of translation) in bacteria. It prevents translocation of the peptidyl tRNA from the A-site to the P-site.
Binding of aminoacyl tRNA to the A site of the ribosome - Inhibited by tetracycline.
Unwinding of supercoiled DNA - inhibited by ciprofloxacin (quinolones), which inhibit DNA gyrase.
Initiation of translation - Inhibited by Streptomycin.
Translocation of peptidyl tRNA - Inhibited by erythromycin.
Peptidyl transferase activity of ribosome - Inhibited by chloramphenicol.
Transcription of mRNA and mRNA synthesis - inhibited by Rifampin (prokaryotes; anti-tuberculosis drug) and alpha amanitin (in eukaryotes)

Question #: 16

A 30-year-old man comes to the physician seeking advice before traveling to a high-altitude location. His vital signs are within normal limits, and his physical examination is unremarkable. To prevent acute mountain sickness, he is prescribed acetazolamide. Which of the following laboratory findings is most likely to be observed a few days after starting this medication?

A. Decreased urine pH; increased arterial pH
B. Increased bicarbonate loss in urine; Increased serum bicarbonate levels
✓C. Increased urine pH; Decreased serum bicarbonate levels
D. Increased urinary acid phosphate and ammonium chloride

E. Increased urine pH; increased arterial pH

Rationale: Acetazolamide inhibits renal carbonic anhydrase. As a result, it inhibits the formation and secretion of H+ in the renal tubular cell. When H+ is NOT secreted into the renal tubule, urinary pH cannot decrease (or urinary pH increases).
Filtered bicarbonate cannot be reabsorbed, and as a result, filtered bicarbonate will be lost in urine, and urinary bicarbonate increases.
When H+ secretion into the renal tubule is inhibited, there would be decreased formation of acid phosphate (titratable acid) and ammonium excretion in urine. The renal tubule is unable to secrete H+, and as a result, it cannot reabsorb filtered bicarbonate, and it cannot form NEW bicarbonate. Serum bicarbonate decreases, which may result in metabolic acidosis and low arterial pH.

Question #: 17

A 23-year-old woman is brought to the hospital by ambulance after suddenly collapsing with muscle weakness and cramps. The patient was participating in an extreme physical event that required sustained maximal exertion for 10 minutes at a high altitude. She has no history of major medical illness. Laboratory studies show rhabdomyolysis, acidosis, and red cell dehydration. Which of the following best explains this patient's findings?

A. Sickle cell anemia
B. Alpha thalassemia
✓C. Sickle cell trait
D. Beta thalassemia
E. Hemophilia A

Rationale: With this question, we wish to highlight that persons who carriers of HbS are, also described as HbAS, under extreme conditions, can show symptoms such as rhabdomyolysis, heatstroke, kidney disease and also have a higher risk of sudden death due to exertion. Extreme conditions which can trigger symptoms can include, but is not limited to, being at high altitudes, performing intense exercise (especially if it leads to dehydration) and high-pressure environments like scuba diving.

Question #: 18

An investigator obtains the whole exome sequence from 565 children born with non-syndromic cleft lip and palate in a search to uncover genetic determinants of this malformation. Which of the following best describes how these data were generated?

A. Automated Sanger sequencing
B. ^{32}-P labeled dideoxynucleotide sequencing
✓C. Next-generation sequencing
D. Comparative genomic hybridization

E. SNP microarray

Rationale: The correct answer is next-gene sequencing which allows massively parallel attainment of sequence data by running millions of sequencing reactions at the same time. Each sequence run is relatively short, and being generated, the sequences are mapped to the previously obtained human genome sequence. In the case of whole exome sequencing, only the approximate 1% of the genome which codes for protein is sequenced.

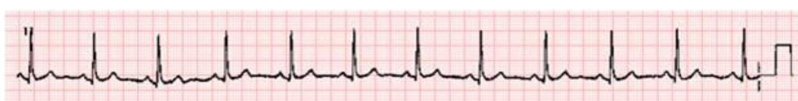
Question #: 19

A 64-year-old man comes to the physician for a routine examination. His vital signs are within normal limits. Physical examination shows no abnormalities. An ECG recorded in lead II is shown in the attached figure. Which of the following ventricular events is most likely occurring during QRS complex in this patient?

A. Opening of the voltage-gated Ca^{++} channel
B. Ca^{++}-induced Ca^{++} release from the sarcoplasmic reticulum
C. Activation of the Na^{+}-K^{+}-ATPase
D. Opening of the voltage-gated K^{+} channel
✓E. Opening of the voltage-gated Na^{+} channel

Rationale: QRS complex indicates ventricular depolarization. Ventricular depolarization occurs when the voltage-gated sodium channels open and sodium influx takes place.

Attachment:



CVS 2.jpg

Question #: 20

A group of medical students watch recording of left ventricular pressure-volume loop from a healthy normal adult. During a particular phase of the cardiac cycle, they observe a sudden rise in the left intraventricular pressure but no change in ventricular volume. Which of the following cardiac valves opens towards the end of this phase?

A. Pulmonary valve
B. Tricuspid valve
✓C. Aortic valve
D. Mitral valve
E. Mitral and aortic valve

Rationale: During isovolumetric ventricular contraction, the ventricular pressure rises without a change in volume. This rise in pressure opens the aortic valve and ejection begins.

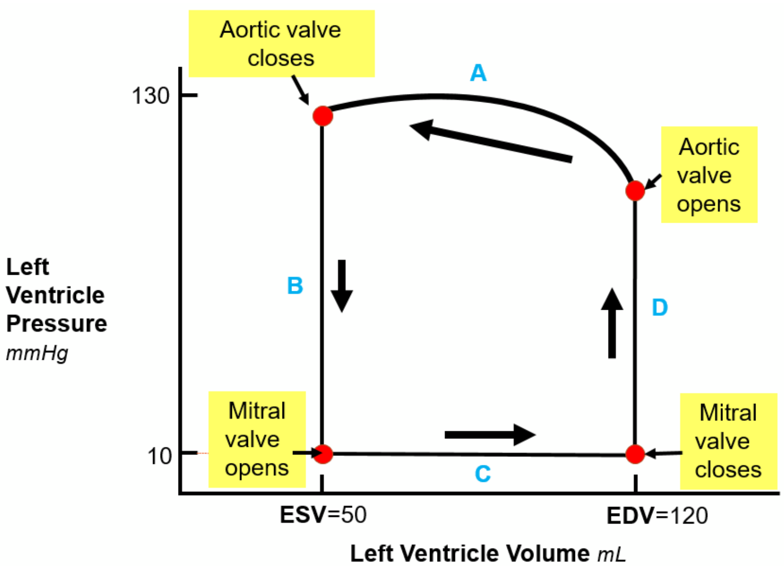
Question #: 21

A 55-year-old man comes to the physician because of a 1-week history of palpitations and dizziness. His blood pressure is 136/86 mm Hg, pulse is 82/min and respirations are 15/min. Physical examination shows a loud S2 in the aortic area. An ECG shows no abnormalities. The pressure volume loop of this patient is shown in the attached figure. Which of the labeled lines in the graph is most likely associated with the third heart sound (S3) sound in this patient?

A. A
B. B
✓C. C
D. D

Rationale: The Pressure-Volume Loop shows the different phases of the cardiac cycle. Line A is related to the phase of ventricular ejection (rapid and reduced). Line B is related to isovolumetric relaxation. Line C is related to ventricular filling (rapid and reduced) and atrial contraction. Line D is related to isovolumetric contraction. S3 sound occurs in early diastole during rapid ventricular filling, which occurs during line C.

Attachment:



Capture.PNG

Question #: 22

A 23-year-old man participates in a study related to the renal function. His vital signs are within normal limits. Physical examination shows no abnormalities. The following test results are obtained from a renal function test:
Urine flow =1.5 ml/min
Plasma inulin =80 mg/ml
Urine inulin =1 mg/ml

Urine osmolality =200 mOsm/kg
Plasma osmolality =300 mOsm/kg
Which of the following values best approximates the patient's free water clearance?

A. 0 ml/min
✓B. 0.5 ml/min
C. 1 ml/min
D. 1.5 ml/min
E. 2 ml/min

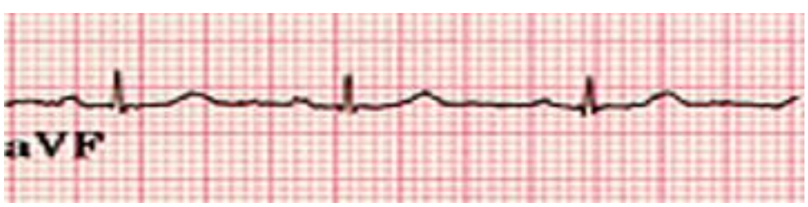
Rationale: CH2O = V - Cosm
Cosm = Uosm x V / Posm = (200) x (1.5)/ 300 = 1 ml/min
CH2O = 1.5 - 1 = 0.5 ml/min

Question #: 23

A 61-year-old woman comes to the physician because of a 2-day history of chest discomfort. She has a 12-year history of migraines treated with a beta-blocker. Blood pressure is 138/82 mmHg, pulse is 82/min and respirations are 12/min. Physical examination is within normal limits. The ECG is shown below. Which of the following ECG lead characteristics most likely corresponds to the recording shown in the attached image?

✓A. Positive electrode located in the left leg
B. Positive electrode located in the left shoulder
C. Positive electrode located in the right shoulder
D. Negative electrode located in the left leg
E. Negative electrode located in the left shoulder

Rationale: The lead aVF is represented in the image. This is a unipolar limb lead, with the positive electrode in the left leg. There is no negative electrode in this type of lead because it is unipolar.
Attachment:



Picture6.png

Question #: 24

A 76-year-old man comes to the physician because of a 4-week history of dizziness and lightheadedness. Two years ago, he had a myocardial infarction. His pulse is 44/min; other vital signs are within normal limits. Physical examination shows no abnormalities. An ECG shows a

PR interval of 0.32 sec (Normal: 0.12-0.2), each QRS complex is preceded by a normal P wave, and a normal QRS interval. Which of the following is most likely?

A. Sinus bradycardia
✓B. First-degree heart block
C. Third-degree heart block
D. Second-degree heart block-type I
E. Second-degree heart block-type II

Rationale: His pulse rate is low, and the PR interval is longer than the normal limit. This is the only ECG abnormality and therefore, he has first degree heart block.

Question #: 25

A 25-year-old woman comes to the physician because of an 8-month history of excessive bleeding during her menstrual cycle. She also feels tired and sleepy and becomes breathless on exertion very quickly. Physical examination shows pale conjunctiva. Her lung function tests are normal. Blood studies show a hemoglobin concentration of 8 g/dL (normal range 12-16 g/dL). Which of the following findings is most likely in this patient?

A. Low arterial PO_2
B. Low Hb O_2 saturation
C. Increased total O_2 content
✓D. Normal arterial PO_2
E. Low P_{50}

Rationale: In anemia hemoglobin(Hb) is low, but affinity of O2 is normal so the saturation is normal. Blood PaO2 (pulmonary arterial & pulmonary capillaries) is 98 - 100mmHg , which is normal. Total O2 content is decreased as most of it is bound to hemoglobin, but the dissolved content is normal. In anemia there is less Hb and no change in P50. Accordingly, the correct choice is 'normal arterial PO2'.

Question #: 26

An experimental substance is being infused intravenously at a constant rate into a healthy volunteer. The substance is known to selectively constrict the efferent arteriole in renal glomeruli. The rate of infusion is closely controlled during the experiment to allow for only mild constriction of the efferent arteriole. Which of the following changes in GFR and filtration fraction (FF) are likely to occur during this infusion?

A. Increase in FF and decrease in GFR
B. FF unchanged and decrease in GFR
C. Increase in GFR and decrease in FF

✓D. Increase in GFR and increase in FF
E. GFR unchanged and increase in FF

Rationale: Constriction of the efferent arterioles increases glomerular capillary hydrostatic pressure. This will increase GFR. Constriction of efferent arteriole increases total renal vascular resistance and therefore decreases renal blood flow as well as renal plasma flow. Since Filtration fraction (FF) is the ratio of GFR to Plasma flow, FF also increases.

Question #: 27

A 49-year-old woman comes to the physician because of a 2-week history of shortness of breath at rest. She was diagnosed with pulmonary fibrosis 3 years ago. Her respirations are 18/min; other vital signs are within normal limits. Physical examination shows a loud second heart sound in pulmonary area, right ventricular heave, and elevated JVP. She is diagnosed with pulmonary hypertension due to her lung disease. Which of the following is most likely increased in this patient?

A. Afterload on the left ventricle
B. Preload on the left ventricle
C. Preload on the right ventricle
✓D. Afterload on the right ventricle
E. Venous return

Rationale: Pulmonary arterial pressure acts as afterload on the right ventricle. In a patient with pulmonary hypertension, the mean pulmonary arterial pressure increases which makes the right ventricle generate higher pressure to pump blood during systole. Increase in afterload on the right ventricle results in increased thickness of the right ventricular wall (right ventricular hypertrophy) as the compensatory mechanism. Increase in right ventricular afterload may indirectly affect the right atrial pressure and reduce venous return. This is the right-sided issue and therefore, left ventricle is not affected in the beginning.

Question #: 28

A 45-year-old man comes to the physician because of a 2-week history of progressive shortness of breath and palpitations. He says he has had one episode of coughing blood. He had several weeks of illness following a sore throat when he was 15-year-old. His pulse ranges 88-126/min with irregular rhythm. Physical examination shows distended jugular venous pulse and rales at the bases of both lung fields. Cardiac auscultation shows a loud first heart sound and a soft diastolic murmur, loudest at the apex. An ECG shows atrial fibrillation. Which of the following valve defects is most likely in this patient?

A. Mitral valve prolapse
✓B. Mitral valve stenosis
C. Aortic stenosis

D. Mitral regurgitation
E. Aortic regurgitation

Rationale: This patient has mitral stenosis. Blood flowing from the left atrium to left ventricle through the stenosed valve produces murmur during diastole (diastolic murmur). Elevated left atrial pressure produces increased pulmonary capillary pressure and pulmonary edema.

Question #: 29

A 69-year-old man comes to the physician because of a 10-month history of progressive shortness of breath on exertion. He does not smoke cigarettes. His vital signs are within normal limits. Physical examination shows no abnormalities. Results of pulmonary function studies are shown:
Vital capacity, FRC, RV and TLC: all reduced,
FEV1/FVC: 83%,
DLCO: Reduced.
Which of the following lung changes is most likely in this patient?

A. Airway obstruction
B. Increased airway resistance
✓C. Decreased lung compliance
D. Increased lung compliance
E. Decreased lung surfactants

Rationale: This patient has a restrictive lung disease because all lung volumes, such as VC, FRC, RV and TLC are reduced. He has a stiff lung and therefore, the lung compliance is decreased.

Question #: 30

A 1-day-old newborn boy is being treated in the neonatal intensive care unit because of respiratory distress. He was born prematurely at 28 weeks' gestation. His pulse is 136/min, respirations are 46/min, and blood pressure is 80/50 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 82% (Normal: 95-98%). With reference to the attached chart, which of the following effects would most likely be associated with this condition?

✓A. A
B. B
C. C
D. D

Rationale: RDS (respiratory distress syndrome) is a syndrome of respiratory difficulty in newborn infants caused by a deficiency of surfactant. The air-fluid interface of the film of water lining the alveoli of the lung (where the exchange of oxygen and CO_2 occurs) exerts large forces that cause the alveoli to close if surfactant is deficient. Lung compliance has decreased, and the work of inflating the stiff lungs is increased. The preterm newborn is further handicapped because his or her ribs are more easily deformed (compliant). Breathing efforts therefore result in deep sternal (breastbone) retractions but poor air entry if the ribs are compliant compared with the lungs. This results in diffuse atelectasis (collapse of the lungs).

Attachment:

	A	B	C	D
Decreased work for alveolar distension		✓	✓	✓
Increased work for alveolar distension	✓			
Small alveoli collapsing into big alveoli	✓			
Big alveoli collapsing into small alveoli		✓	✓	✓
Atelectasis	✓		✓	
Emphysema		✓		✓

RespDistress.jpg

Question #: 31

A 43-year-old woman volunteers for a study in a respiratory clinic. Her pulse is 80/min, and her blood pressure is 130/70 mm Hg. Physical examination show lungs are clear on auscultation and no cardiac abnormalities. After the evaluation, her pulmonary function and blood flow is measured. Blood flow to all regions of her lungs is considered to be in Zone III. This change is most likely caused by which of the following?

A. Positive pressure ventilation
B. Severe dehydration
✓C. Moderate exercise
D. Exposure to high gravitational forces
E. Significant blood loss

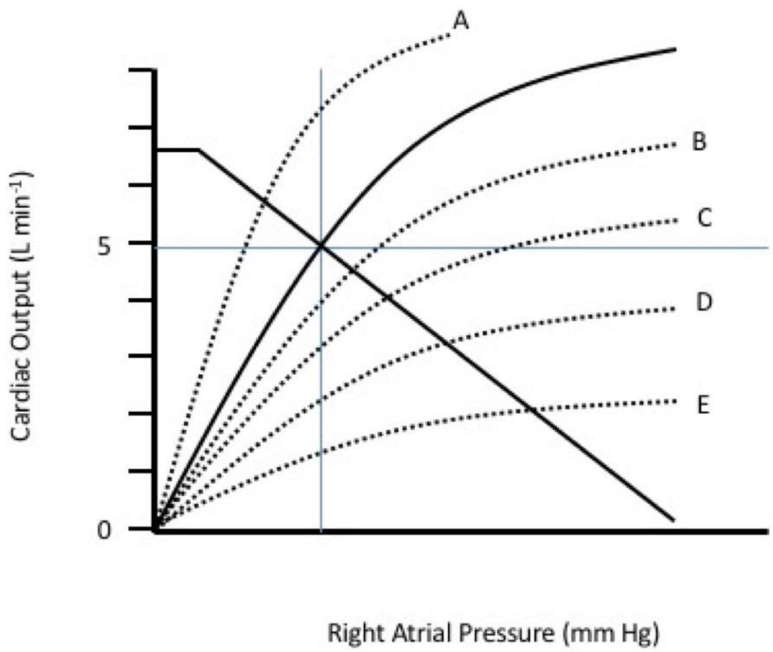
Rationale: This concerns lung zoning, and we assume that the patient is healthy. Arterial blood pressures in the alveolar blood vessels to different parts of the lungs changes vertically, with increased pressures at the bottom of the heart. The zones compare the changes in the arteriolar-venous pressures (which change) in comparison to the Alveolar pressures (which do not change). Zone 1 is non-physiological and can occur with reduced perfusion or forced ventilation, and Zones 2 and 3 occur normally throughout the lungs, with zone 2 higher up in the lungs that zone 3. With increased perfusion, blood flow and pressures to the lungs increases converting zone 2 regions to zone 3.

Question #: 32

A 61-year-old woman comes to the physician because of a 5-week history of fatigue and breathlessness on exertion. She has a 5-year history of coronary artery disease. Her pulse is 95/min and blood pressure is 142/86 mmHg. Physical examination shows no abnormalities. She undergoes an invasive procedure for the measurement of cardiac output, and the following values are determined:
Total oxygen consumption =210 ml/min,
Arterial-venous oxygen difference =0.07 ml/ml blood.
A series of cardiac performance curves (solid lines for normal) are shown in the attached figure. Assuming the venous return curve is normal, which of the following cardiac function curves is most likely in this patient?

A. A
B. B
C. C
✓D. D
E. E

Rationale: This is a 2-step question. First you have to determine cardiac output (CO) using the equation from Fick's principle and then determine which cardiac performance line gives the calculated CO. Fick's principle is   cardiac output (CO) =   Oxygen consumption/arteriolar-venous O2 difference.. So, the CO would be 210/0.07 = 3000 mL or 3 Liters. You will have to then determine which of the cardiac performance lines cross the venous function curves at a CO (y-axis) of 3L. The only one that is correct is Curve D.
Attachment:



cardiac performance curv. pg

Question #: 33

A 72-year-old woman comes to the physician because of a 3-day history of productive cough and increased breathlessness. She has no fever or chest pain. She has a 40 pack-years history of cigarette smoking and an 8-year history of worsening respiratory problems with shortness of breath at rest. Her respirations are 18/min with shallow tidal volume. Physical examination shows labored breathing with accessory muscle use on expiration and pursed lip breathing. She is diagnosed with obstructive lung disease. Which of the following is most likely increased in this patient?

A. Vital capacity
B. Forced vital capacity
✓C. Total lung capacity
D. Forced expiratory volume in 1 second
E. Inspiratory capacity

Rationale: The use of accessory muscles of expiration indicates a disease primarily affecting the expiratory phase of breathing. Obstructive pulmonary diseases primarily affect the expiratory phase of respiration. The pursed lip breathing along with the productive cough in a smoker indicate that this patient is most likely to have pulmonary emphysema. In emphysema the total lung capacity increases because air is obstructed from leaving the lung. The vital capacity forced vital capacity and forced expiratory volume in 1 second is therefore decreased.

Question #: 34

A 60-year-old man comes to the physician because of a 1-month history of progressive shortness of breath. He has a 23-pack-year smoking history. His respirations are 24/min, and his pulse is 102/min. Physical examination shows the use of accessory muscles during expiration, excessive coughing, pursed lip breathing, and bilaterally decreased breath sounds. An x-ray of the chest shows hyperinflated lungs. Which of the following changes is most likely in this patient?

A. Decreased compliance
✓B. Decreased elastic recoil
C. Increased surfactant
D. Increased gas exchange
E. Increased ventilation

Rationale: Based on the clinical presentation the patient likely has a pulmonary disease that causes increased compliance (hyperinflated lungs, increased expiratory effort, pursed lip breathing). Compliance and elastic recoil (elastance) are inversely proportional. Therefore, this patient will most likely have decreased elastic recoil. This condition is likely to cause decreased ventilation and gas exchange.

Question #: 35

A 35-year-old man comes to the physician because of shortness of breath of 6-month duration. He has 10 pack year history of cigarette smoking. He is advised to undergo pulmonary function tests. The spirometry results show:
Total lung capacity =6000 mL.
Expiratory Reserve Volume =1000 mL.
Inspiratory Reserve Volume =3300 mL.
Tidal Volume =500 mL and
Vital Capacity =4800 mL.
Calculate the functional residual capacity (FRC) of this patient?

A. 1200 mL
B. 1500 mL
C. 2000 mL
✓D. 2200 mL
E. 2500 mL

Rationale: TLC=6000; VC=4800.
Therefore, Residual volume, RV=TLC-VC = 6000-4800=1200.
FRC = RV + Expiratory reserve volume = 1200+1000 = 2200.

Question #: 36

A 22-year-old woman comes to the physician because of a 6-week history of headache, blurred vision, and fatigue. Her blood pressure is 180/110 mmHg. Physical examination shows no abnormalities. Laboratory studies show elevated serum aldosterone, undetectable serum renin, hypernatremia, hypokalemia, elevated serum bicarbonate, and a blood pH of 7.52. She is diagnosed with primary hyperaldosteronism. Which of the following renal tubular changes is most likely in this patient?

✓A. Increased activity of epithelial Na+ channels in principal cells
B. Decreased activity of H+/K+ antiporter in α-intercalated cells
C. Decreased activity of Na+/K+/2Cl- cotransporter in the loop of Henle
D. Decreased activity of Na+/H+ antiporter in the proximal convoluted tubules
E. Increased activity of Na+/K+-ATPase in the proximal convoluted tubules

Rationale: Aldosterone acts on the principal cells of the collecting ducts and increase activity of the epithelial sodium channels.

Question #: 37

A 60-year-old man is brought to the physician because of a 1-hour history of shortness of breath. He has a 2-year history of deep vein thrombosis. His respirations are 24/min with increased tidal volume. Pulmonary arteriogram shows block in the segmental branches of the pulmonary artery because of thromboembolism. Arterial blood gas studies show a partial pressure of arterial blood oxygen (PaO2) of 64 mm Hg (Normal: 85-95 mm Hg). Which of the following mechanisms best explains this patient's hypoxemia?

A. Decreased lung surfactants
B. Decreased PO2 in inspired air
C. Hypoventilation of central origin
D. Pulmonary edema
✓E. Ventilation-perfusion mismatch

Rationale: This patient has pulmonary embolism which blocks blood flow to certain segment of lung and produce a dead space. As the blood flow to this region of the lungs is blocked, this creates a mismatch between ventilation and perfusion and hence, V/Q mismatch.

Question #: 38

A 79-year-old woman comes to the emergency department because of a 4-hour history of breathlessness. She has a 13-year history of progressive cough and breathlessness. She has smoked 1 pack of cigarettes daily for 30 years. She weighs 70 kg (154 lbs) and is 170 cm (5 ft 7 in) tall. Her temperature is 38.2 °C (101 °F), pulse is 112/min, respirations are 26/min. Physical examination shows diminished respiratory sounds and wheezing bilaterally. Cardiac auscultation shows no abnormalities. An x-ray of the chest shows hyperinflation that indicates emphysema. Which of the following additional x-ray findings is most likely in this patient?

A. Air under the diaphragm
✓B. Flat diaphragm
C. Mediastinal shift
D. Elevated hemidiaphragm
E. Atelectasis

Rationale: A patient with emphysema has a large residual volume and FRC. His lungs are overinflated which pushes the diaphragm down. This is observed in chest x-ray as a flat diaphragm.

Question #: 39

A 40-year-old man comes to the physician because of a 2-week history of nausea, vomiting and abdominal pain. Physical examination shows no abnormalities. Gastric biopsy specimens obtained from endoscopy show a patchy erosion of the epithelial lining with mucosal infiltration by lymphocytes, macrophages, and plasma cells. Which of the following best describes this type of inflammation?

A. Acute suppurative inflammation
B. Acute fibrinous inflammation
C. Acute serous inflammation
✓D. Chronic inflammation
E. Fibrosis

Rationale: The chronic inflammatory infiltrate is characterized by the large numbers of lymphocytes, macrophages, and plasma cells in contrast to acute inflammation where the major cell type is the neutrophil. Chronic granulomatous inflammation is characterized by the clusters of epithelioid macrophages and multinucleate giant cells.

Question #: 40

A 32-year-old woman comes to the physician for a follow up examination. She received a stab wound in her abdomen 3 weeks ago. Physical examination today shows that the sides of the wound are progressing well toward each other. The physician notes that this action is contributed to by the influence of a certain junctional complex that links the cell's actin cytoskeleton to the extracellular matrix. Which of the following transmembrane protein is a component of this junctional complex?

A. Selectin

B. Cadherin
✓C. Integrin
D. Desmocollin
E. Claudin

Rationale: Both focal adhesions and hemidesmosomes are strong anchoring junctions located on the basal surface of epithelial cells. Integrins are the main family of proteins involved in both focal adhesions and hemidesmosomes that bind to extracellular glycoproteins such as fibronectin and laminin, respectively.

Question #: 41

A 62-year-old woman comes to the physician because of a 2-week history of cough and fever. Physical examination shows a temperature of 38°C (100.4°F) and wheezing breath sounds. X-ray of the chest shows infiltration of the inferior lobes of both the right and left lungs. She is diagnosed with bilateral bronchopneumonia. In this disease, the respiratory portion of the lungs become filled with inflammatory exudate. Which of the following is a characteristic feature of this portion of the respiratory tract?

A. Cartilage
B. Mucous glands
C. Goblet cells
D. Ciliated columnar cell
✓E. Alveoli

Rationale: The respiratory portion is the part of the respiratory tract in which alveoli are found, and gas exchange occurs.

Question #: 42

An 85-year-old woman is brought to the emergency department by her son because of a 4-hour-history of severe right hip pain and deformity, after falling out of bed. Physical examination shows right hip tenderness and external rotation of the right lower limb. An x-ray of the pelvis and hip shows a fracture in the neck of the right femur. Bone mineral density measurement shows severe osteoporosis. Which of the following cells are most likely involved in the disease process seen on bone mineral density measurement?

A. Chondrocytes
B. Chondroblasts
✓C. Osteoclasts
D. Fibrocytes
E. Fibroblasts

Rationale: Osteoporosis is characterized by the progressive loss of bone density caused by increased activity of osteoclasts and bone resorption. This results in decreased bone mass, enhanced bone fragility, and increased risk of bone fracture.

Question #: 43

A 75-year-old woman is brought to the emergency department by her husband because of a painful and deformed left hip after falling out of bed 4-hours ago. Physical examination shows left hip tenderness and external rotation of the left lower limb. An x-ray of the pelvis and hip shows fractures of the neck and head of the left femur. Bone mineral density measurement shows severe osteoporosis. Which of the following best characterizes the cartilage most likely to be affected in this disease process?

A. Imparts tensile strength to tissues
B. Presence of elastic fibers
✓C. Rounded chondrocytes in the intermediate zone
D. Hypertrophied chondrocytes in the zone of hypertrophy
E. Readily regenerates

Rationale: Osteoporosis is characterized by the progressive loss of bone density caused by increased activity of osteoclasts and bone resorption. There is also marked destruction of the articular cartilage, located at the ends of the epiphyses, lining the joint spaces. This cartilage also gets eroded and destroyed during the disease process. The articular cartilage is made up of 4 zones (superficial/tangential, intermediate/transitional zone - option c, deep/radial zone, and calcified zone).

Question #: 44

A 21-year-old man is brought to the emergency department because of a 1-hour history of injury to his left ankle during a football match. Physical examination shows a swollen, tender left ankle with reduced active range of motion. An MRI shows a rupture of one of the ankle ligaments. Which of the following types of connective tissue most likely forms the ruptured structure?

✓A. Dense regular
B. Loose
C. Adipose
D. Dense irregular
E. Mucous

Rationale: Ligaments and tendons are responsible for resisting unidirectional stress, therefore the fibers within these structures are arranged in a parallel array and are densely packed (Dense Regular CT) to provide maximum strength.
Option B Loose CT is found beneath most epithelial linings.
Option D: Dense irregular CT is found in organ capsules, periosteum, dermis
Option E Mucous CT is found in the umbilical cord, cardiac jelly, and vitreous humor of the eye.

Question #: 45

Investigators are developing a new drug to treat connective tissue disorders and aim to study its effects on connective tissue elements in the body. The drug is given intravenously, and x-rays are taken of different sites at various times. They discover that the drug has a preference for bony tissue. Biopsy samples obtained from some participants show destruction of specific connective tissue elements within the bone. Which of the following connective tissue fibers is most abundant in the affected tissue?

A. Type III collagen
B. Elastic
C. Type IV collagen
D. Type II collagen
✓E. Type I collagen

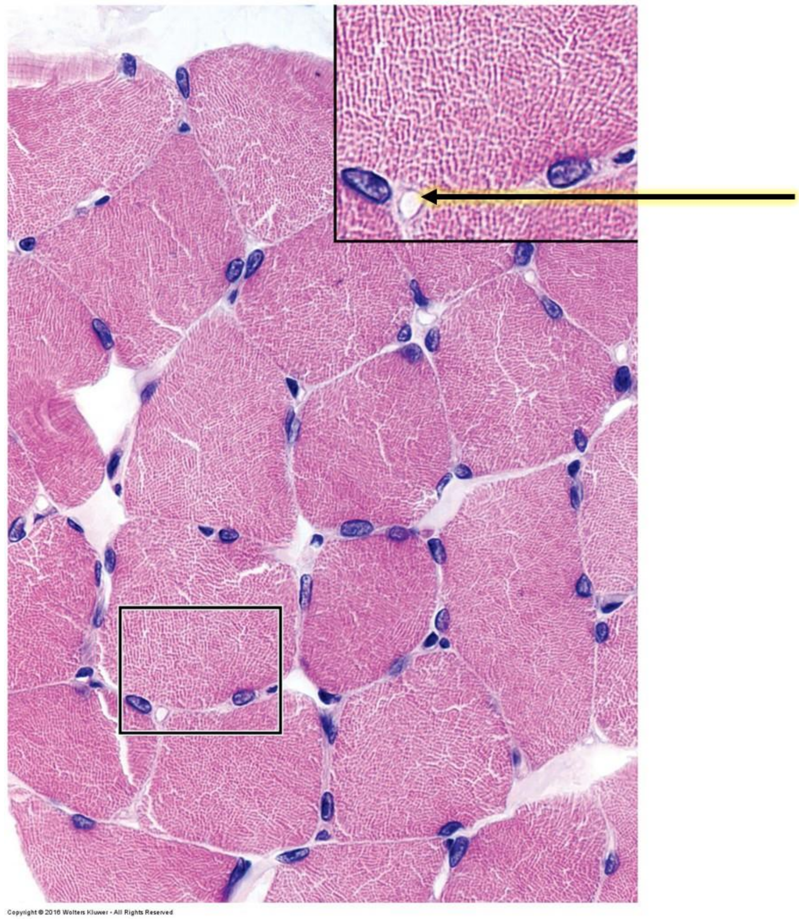
Rationale: Bone is specialized connective tissue, and as such is made up of cells and an extracellular matrix. That matrix is composed of fibers and ground substance. The predominant fiber in bone is type I collagen.
Type II is found mainly in cartilage and type iv in basement membrane while type III is more in areas of loose cells, forming a structural support meshwork such as in the immune system.

Question #: 46

A 25-year-old woman comes to the emergency department because of a 6-month history of pain to her right leg. Physical examination shows a 5.2cm x 3cm solid mass involving the muscles of her right thigh. Biopsy shows increased nuclear to cytoplasmic ratio with nuclei fragmentation and destruction of the capillaries highlighted at the tip of the arrow. Which of the following best corresponds to these capillaries?

A. Sinusoidal
B. Arteriole
C. Fenestrated
D. Venule
✓E. Continuous

Rationale: The image shows skeletal muscle which (like most tissues) is supplied by continuous capillaries. These vessels are characterized by the endotheliocytes joined by numerous tight junctions and lying on a continuous basal lamina.
Attachment:



continuous capillary .jpg

Question #: 47

A 46-year-old man comes to the emergency department because of a 2-week history of a painful ulcer on his right foot. he has a history of poorly controlled Type 2 diabetes mellitus. His temperature is 40°C (104°F). Physical examination shows a 6-cm, foul-smelling, pus-filled ulceration on his right distal foot with inflammation of the surrounding skin. He is diagnosed with a necrotizing infection. Which of the following best characterizes the cell death responsible for these findings?

A. Formation of membrane bound vesicles
B. Aggregation of chromatin
C. Fragmentation of nucleus
D. Plasma membrane blebbing
E. Cell shrinkage
✓F. Cell swelling

Rationale: Apoptosis and necrosis represent the 2 mechanisms of cell death. Both processes carry distinct morphological and biochemical differences. Apoptosis is the regulated mechanism of the two, and it proceeds in an orderly way, and includes features of all options listed except for cell swelling, which is more characteristic of necrosis.

Question #: 48

A 37-year-old woman comes to the physician because of a 1-month history of swelling on the medial aspect of her arm. The most probable diagnosis after examination is a lipoma, a benign tumor of adipocytes. A biopsy is collected for histopathological examination. Which of the following is most likely to be used to stain the lipid in these cells for microscopic study?

A. Hematoxylin
B. Eosin
C. Silver
✓D. Oil red O
E. Periodic acid Schiff

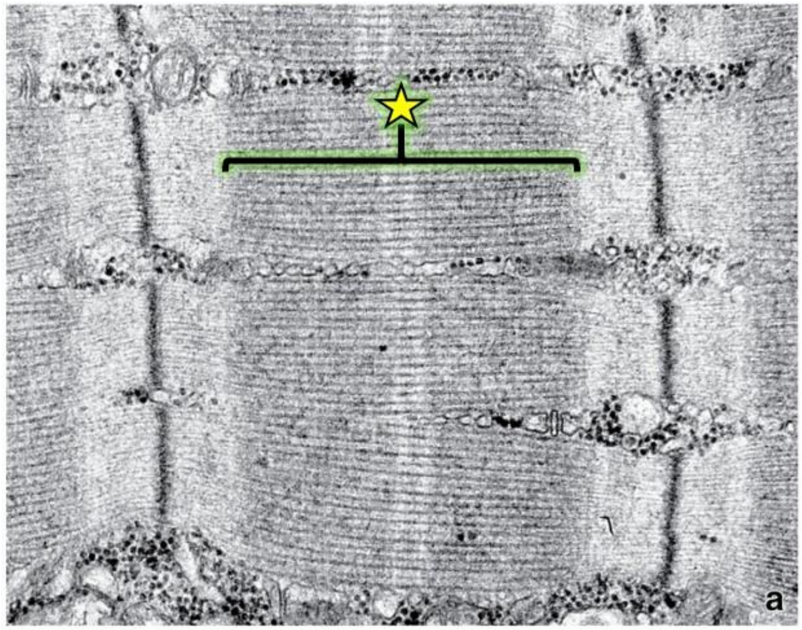
Rationale: Oil red O is a special stain for lipids. It is performed on fresh samples as alcohol fixation removes most lipids.

Question #: 49

A 32-year-old man is brought to the emergency department because of a 1-hour history of recurrent seizures. History shows he is an epileptic person. Physical examination shows tonic-clonic seizures with spasm of the skeletal muscles. Which of the following best characterizes the structure indicated on the attached micrograph?

A. Presence of thin filaments only
B. Presence of thick filaments only
C. Absence of any kind of filaments
✓D. Presence of both thick and thin filaments
E. Found abundantly in smooth muscle cells

Rationale: The A band is characterized as an area where there is overlap of thin and thick filaments.
Attachment:



A band.jpg

Question #: 50

A 64-year-old man comes to the physician because of a 2-week history of swelling to both his feet. He has a history of diabetes mellitus (type II) and hypertension diagnosed 30 years ago. Laboratory studies of blood show elevated glucose concentration. Urinalysis shows high concentration of glucose. Which of the following structures most likely is the site of reabsorption of the elevated substance found on urinalysis?

A. Thick ascending limb of the loop of Henle
B. Distal convoluted tubules
C. Descending thin limb of the loop of Henle
D. Collecting ducts
✓E. Proximal convoluted tubules

Rationale: The proximal tubule is the initial and major site of reabsorption & reabsorbs nearly all glucose, amino acids, and small polypeptides.